1.0 Entity-Relationship Diagram (ERD)

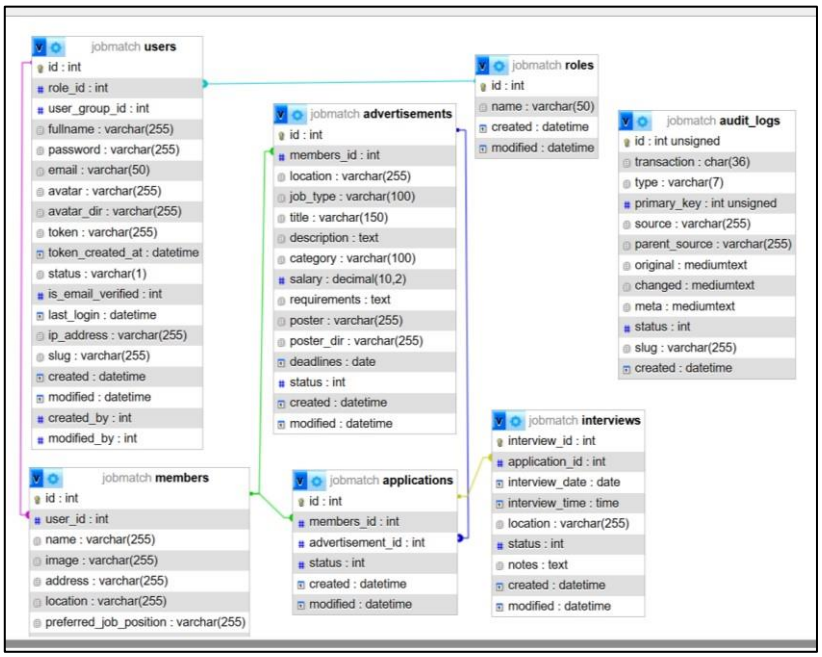


Figure 1: JobMatch's ERD

This system consists of the following main components: Users, Members, Roles, Advertisements, Applications, Interviews, and Audit Logs. Each entity holds particular data which is needed for the efficient management of the system. For instance, the Users entity holds login details along with account information, whereas the Members entity contains extra personal information for the users (job seekers) as well as the admins. The access level for users is defined at the Role entity level, which allows for the appropriate levels of access for the administrators and users

(job seekers) in the system.

The relations among these entities help realise the JobMatch system's core business processes. Each user gets assigned one role, which gives this user a profile with detailed personal information. Admins may publish multiple advertisements, and users (job seekers) may post multiple applications to various advertisements. The Interview entity is used with the recruitment workflow to schedule and manage interviews of applicants. It keeps interview information, including interview date, interview time and interview status, making the interview process well managed and monitored during the hiring procedure. The Audit Logs entity also logs any user activity, including login attempts and system activity, which can help you track users, bolster security, and help with system monitoring.

2.0 System Requirements

2.1 Software

Software	Version	Primary Function
Laragon	V8.6.1	Local Development Environment: Provides Apache, PHP, and MySQL services to run the JobMatch system locally during development and testing.
PHP	PHP 8.3.30	Server-side scripting Language: Eecutes the application logic and process user requests in the JobMatch system.
MySQL	MySQL 8.4.3	Database Management System: Stores and manages application data such as users, job advertisements, applications, and members.
phpMyAdmin	V5.2.3	Database Administration Tool: Used to create, import, manage, and maintain the MySQL database through a web interface.
Visual Studio Code	1.128.0	Integrated Development Environment (IDE): Used to develop, edit, debug, and manage the project source code.
Google Chrome	Latest version	Web Browser: Used to access, test, and verify the functionality of the JobMatch web application.

Table 1: System Requirements

2.2 Version Compatibility

Component	**Compatible Version**	**Compatibility Description**
CakePHP	5.2	Compatible with PHP 8.1 or above and used as the main framework for developing the JobMatch system.
PHP	8.3.30	Fully compatible with CakePHP 5.2 and supports the required server-side functionalities.
MySQL	8.4.3	Compatible with CakePHP through PDO and used to store and manage application data.
Apache	2.4.66	Supports PHP 8.3.30 and is used to host the JobMatch application locally.
Laragon	8.6.1	Provides a compatible environment by integrating Apache, PHP, and MySQL for local development.
phpMyAdmin	5.2.3	Compatible with MySQL 8.4.3 for managing the project database through a web interface.
Google Chrome	Latest Version	Fully supports HTML5, CSS3, and JavaScript features for testing the web application.

Table 2: Version Compatibility

2.3 Dependencies

The Jobmatch application uses Composer to manage its dependencies. The main packages required by the system are listed below.

Dependencies	**Version**	**Primary Function**
CakePHP	5.2	PHP Framework: Provides the MVC architecture, routing, controllers, models, and views for developing the JobMatch web application efficient.
Bootstrap	v5.x	Front-end framework: Used to create responsive, mobile-friendly, and modern user interface components.
Composer	2.10.2	Dependency Manager: Installs and manages CakePHP packages and third-party libraries required by the system.

Table 4.3: Dependencies

3.0 User Interface Overview

The JobMatch system features a modern, clean, and responsive user interface designed to provide an excellent user experience on both desktop and mobile devices.

3.1 Overall Layout Structure

The system uses a consistent layout throughout all pages with the following components:

3.1.1 SideBar

1. **System Logo:** Display the official circular JobMatch UiTM logo at the very top.
2. Main Navigation Links: A clean list of active navigation links accompanied by a visual icon:

   (User Interface)

   - Dashboard
   - Profile
   - Application

   (Admin Interface)

   - Dashboard
   - Profile
   - Advertisement
   - Users
   - Audit Logs

3. **Logout Button:** Positioned at the bottom of the sidebar.

3.1.2 Header

The Header is a clean, responsive top navigation bar that is composed of the slide toggle (hamburger icon) on the far left, which expands/collapses the sidebar, and on the far right, it contains the crescent moon icon to switch between dark and light versions.

3.1.3 Main Content Area

The main content area contains dynamic user dashboard info that comes with a very neat card-and-grid arrangement. It is launched with a personalised greeting banner welcome user by Welcome back, Hawa (User). Immediately below is a search bar and filter bar that contains a text box that shows job titles, a dropdown that says All Status and a large golden-yellow search button.

Further down are four metric cards to give a glance at application data, that has shown that the user has applied a total of 1, there is 1 pending, 0 approved, and 0 rejected.

After these statistics are two quick access buttons, which are Apply for a Job and a white coloured button that simply reads Update Profile and has a clean dark border. Below is a two-column grid that has a recent applications table (job position, date and status, a yellow-coloured pending badge) in the first column, while a section highlighting the latest available job openings in the other column, such as Barista at Cafe Jasmine, UiTM Puncak Perdana, and an Apply button for that job opening.

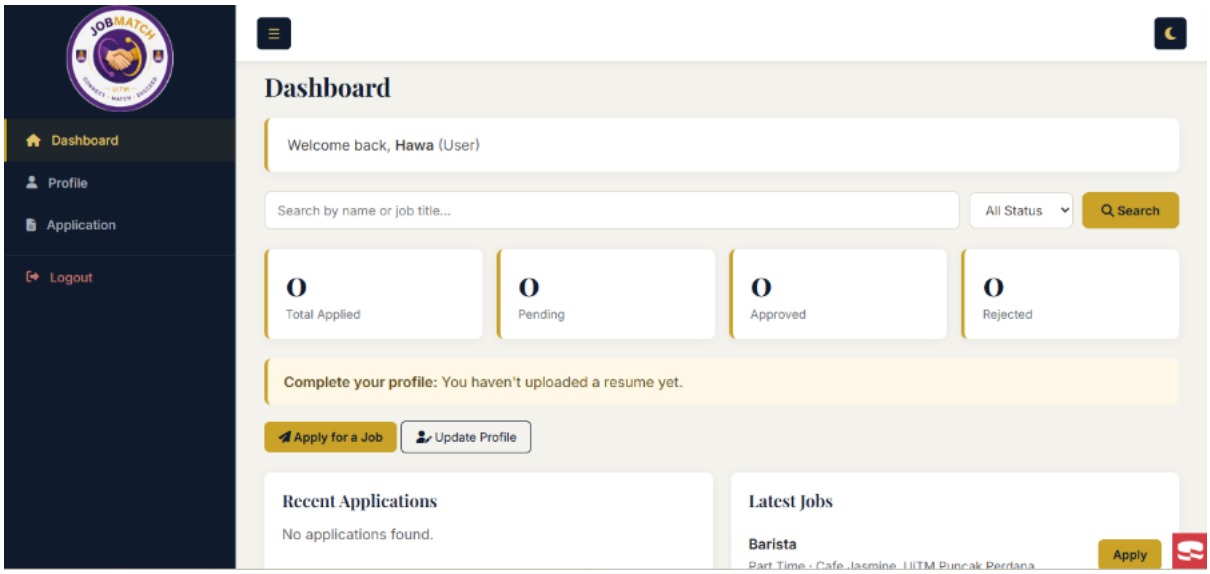


Figure 2: Layout JobMatch

3.2 Navigation System

Applicant Navigation Menu:	**Administrator Navigation Menu:**
Dashboard	Dashboard
Profile	Profile
Application	Advertisement
Logout	Users
	Audit Logs
	Logout

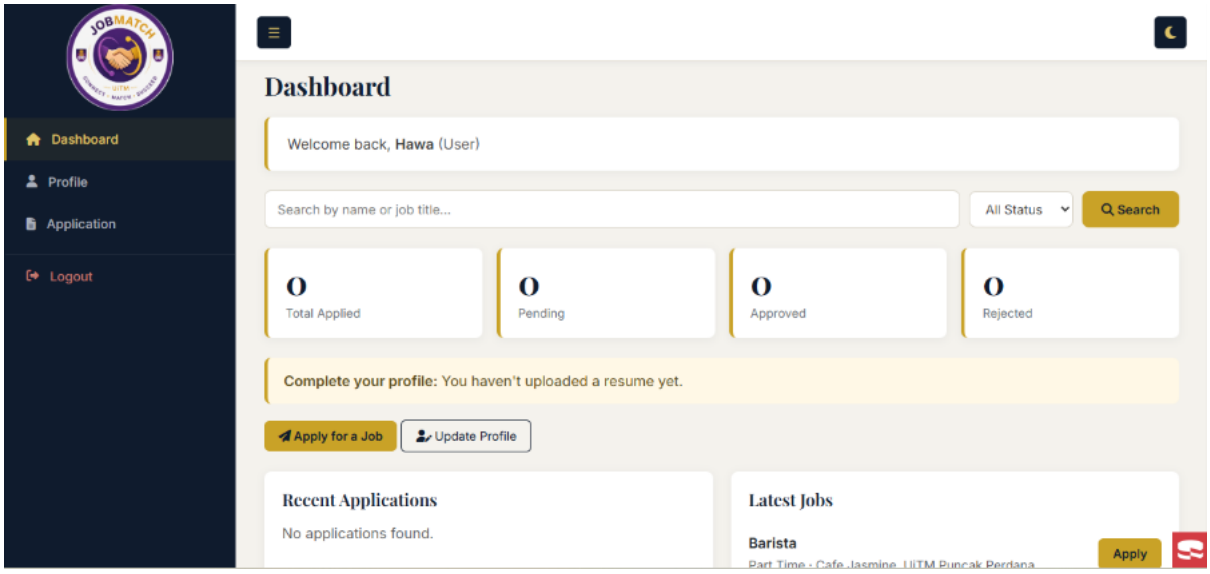


Figure 3: Navigation bar for applicant

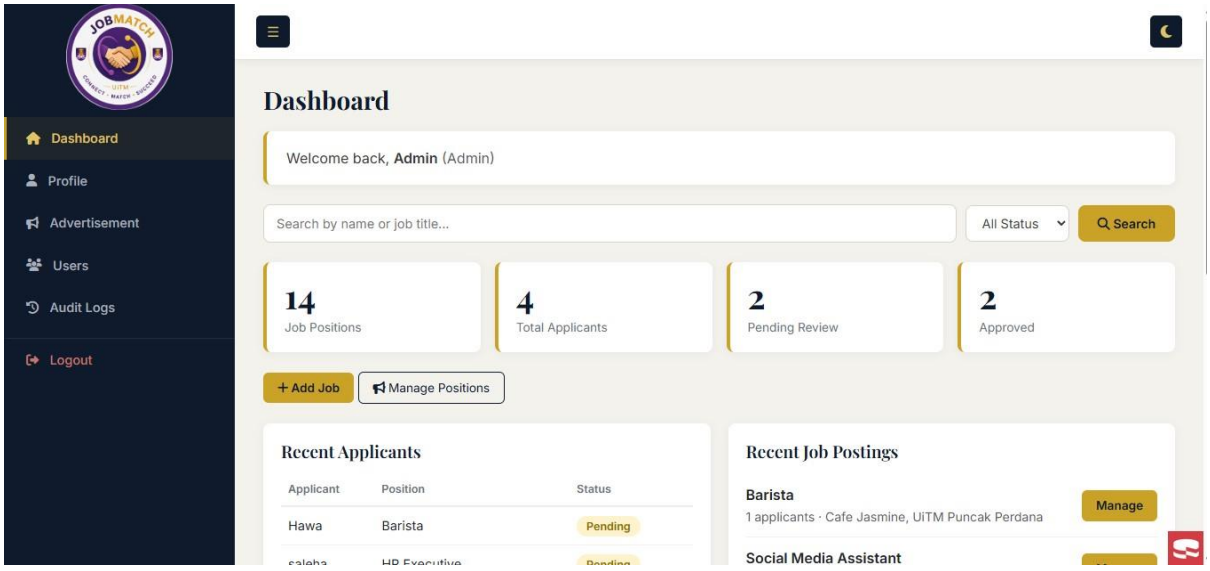


Figure 4: Navigation bar for admin

4.0 Features & Functionalities

4.1. Interactive Job Advertisement



Figure 5: JobMatch Advertisement Page

The Interactive Job Advertisement function serves as the primary access point of the JobMatch website, where users can review available positions before registering. The advertisements have a brief description of the job summary, which includes the name of the job, the company that offers the job, and its location. Once users click a job advertisement, they can view the job description, which highlights responsibilities, required qualifications, salary details (if available), and the job deadline.

4.2. Smart Job Search

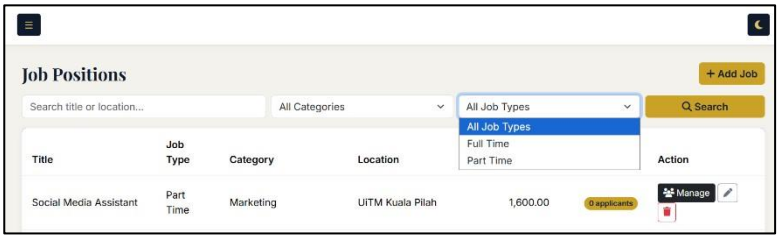


Figure 6: Job Search 1

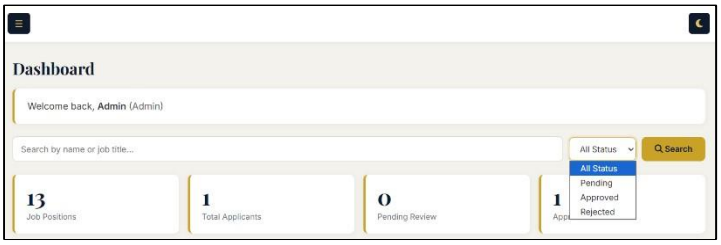


Figure 7: Job Search 2

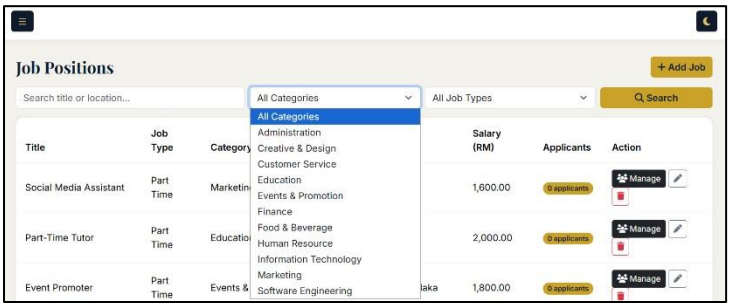


Figure 8: Job Search 3

Smart Job Search features is aimed in helping users to find appropriate jobs effectively without searching through all job vacancies available one by one. Users may look for jobs by entering keywords such as job title, company name, or any necessary skills. Thus, users will be able to get their desired jobs in a faster way. All search results will appear instantly, enabling users to make comparisons between various job vacancies instantly with their preferences. This functionality considerably enhances the entire user experience, making job finding faster and more comfortable for users.

4.3. Resume Upload

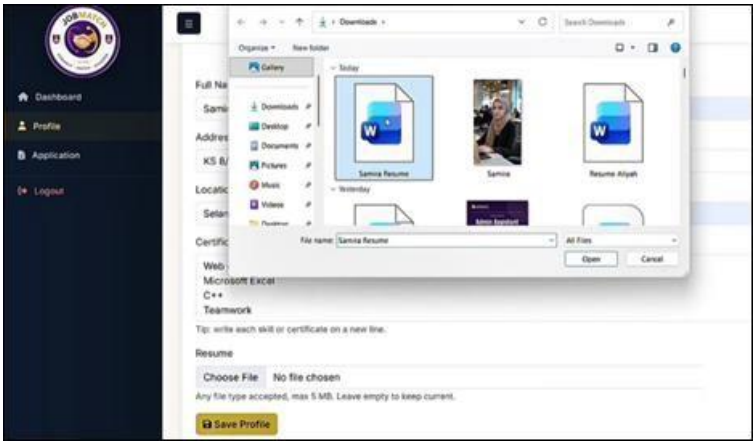


Figure 9: Resume Upload

Resume Upload features make the application process easier by helping people upload and maintain their resumes using the profile page itself. Once the resumes have been uploaded, the person does not need to provide all the same information every time they want to apply for different job vacancies. It is a quick way to make applications, and at the same time, makes sure that employers get the latest version of the resume of an applicant. In reducing repeated tasks, the system promotes user to apply for more options and makes the entire recruitment process more efficient.

4.4. Admin Recruitment Dashboard

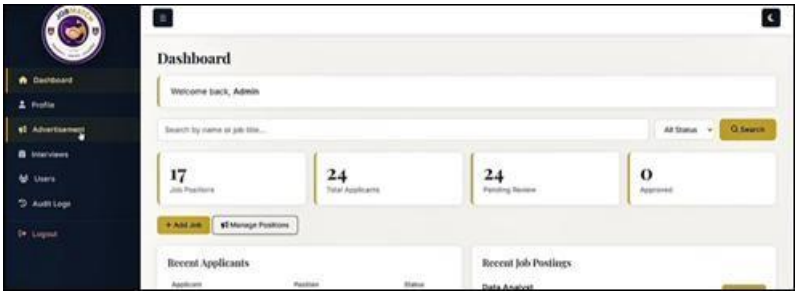


Figure 10: Admin Dashboard

The Admin Recruitment Dashboard gives a detailed insight into all the recruitment processes conducted within the JobMatch website. This tool presents as vital information that show the total amount of received job applications, vacant job posts, applicants' activity, and system statistics. At the same time, administrators can search for particular records, create new job advertisement posts, and control all vacant jobs from one screen. In this way, the administration can control the hiring process and make quick decisions, when necessary, through the presentation of recruitment information on a dashboard.

5.0 Workflow of Form

Below is the breakdown of the step-by-step workflow in the job matching system from the beginning of the process (user registration) to the end of the system process (job application result): It involves the cooperation between the user (applicant) and the system administrator.

Here is the detailed breakdown of each step in the workflow:

